METRIC

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MILITARY SPECIFICATION

PURIFIERS, SOLIDS EJECTING (SELF CLEANING), CENTRIFUGAL, FUEL AND LUBE OIL

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the construction, performance, and delivery acceptability of centrifugal, solids ejecting (self cleaning), lubricating oil and fuel purifiers for service on board Navy ships.

1.2 Classification. Centrifugal purifiers shall be of the standard sizes specified in table I (see 6.2).

TABLE I. Purifier size and minimum capacity.

Purifier size	Capacity		
	F-76 fuel at 21°C (50°F) (MIL-F-16884)	Diesel engine lube oil at 79°C (175°F) (MIL-L-9000)	Turbine lube oil at 71°C (160°F) (MIL-L-17331)
SC-1	2650 (700) 1/	628 (166) 1/	1892 (500) 1/
SC-3	416 (110) 2/	121 (32) 2/	371 (98) 2/

1/ Capacity in liters per hour (gallons per hour).
2/ Capacity in liters per minute (gallons per minute).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 05Q42, 2531 Jefferson Davis Hwy., Bldg. 3, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

FF-B-171	- Bearings, Ball, Annular (General Purpose).
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
QQ-S-763	- Steel Bars, Wire, Shapes, and Forgings, Corrosion Resisting.

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MIL-S-901	- Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-S-1222	- Studs, Bolts, Hex Cap Screws, Socket Head Cap Screws and Nuts.
MIL-C-2212	- Contactors and controllers, Electric Motor A.C. or D.C., and Associated Switching Devices.
MIL-S-5000	- Steel, Chrome-Nickel-Molybdenum (E4340) Bars and Reforging Stock
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4, JP-5 and JP-5/JP-8 ST.
MIL-L-9000	- Lubricating Oil, Shipboard Internal Combustion Engine, High Output Diesel.
MIL-P-15024	- Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5	- Plates, Identification.
MIL-S-16032	- Switches and Detectors, Shipboard Alarm Systems.
MIL-H-16313	- Heaters, Fluid, Fuel Oil, Naval Shipboard.
MIL-P-16789	- Pumps (Including Prime Movers and Support Items); Packaging of.
MIL-F-16884	- Fuel, Naval Distillate.
MIL-M-17060	- Motors, 60-Hertz, Alternating Current, Integral-Horsepower, Shipboard Use.
MIL-I-17244	- Indicators, Temperature, Direct-Reading, Bimetallic, (3 and 5 Inch Dial).
MIL-L-17331	- Lubricating Oil, Steam Turbine and Gear, Moderate Service.

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MIL-M-17413 - Motors, Direct Current, Integral H.P., Naval Shipboard.
MIL-H-17428 - Heaters, Fluid, Industrial (Lubricating Oil) (For Naval Shipboard Use).
MIL-M-17508 - Mounts, Resilient: Types 6E100, 6E150, 7E450, 6E900, 6E2000, 5E3500, 6E100BB, 6E150BB, 7E450BB and 6E900BB.
MS17828 - Nut, Self-Locking, Hexagon, Regular-Height, (Nonmetallic Insert) 250°F, Nickel-Copper Alloy.
MS17829 - Nut, Self-Locking, Hexagon, Regular Height, 250°F, (Non-Metallic Insert) Non-Corrosion-Resistant Steel.
MS17830 - Nut, Self-Locking, Hexagon-Regular Height, 250°F, (Non-Metallic Insert) 300 Series Corrosion Resistant Steel.
MIL-B-17931 - Bearings, Ball, Annular, For Quiet Operation.
MIL-G-18997 - Gauge, Pressure, Dial Indicating.
MIL-P-19131 - Pumps, Rotary, Power Driven, Miscellaneous.
MIL-L-19140 - Lumber and Plywood, Fire-Retardant Treated.
MIL-S-24093 - Steel Forgings, Carbon and Alloy, Heat Treated.
MIL-H-24299 - Heaters, Lubricating Oil, Electric, Naval Shipboard.
MIL-P-24441 - Paint, Epoxy-Polyamide, General Specification for.
MIL-P-24441/1 - Paint, Epoxy-Polyamide, Green Primer, Formula 150, Type I
MIL-P-24441/2 - Paint, Epoxy-Polyamide, Exterior Topcoat, Haze Gray, Formula 151, Type I
DOD-F-24669/6 - Forgings and Forging Stock, Steel Bars and Billets, Corrosion Resisting; for Reforging. (Metric)
MIL-P-24691/3 - Pipe and Tube, Corrosion-Resistant, Stainless Steel, Seamless or Welded.
MIL-C-24707/1 - Castings, Ferrous, For Machinery and Structural Applications.
MIL-C-24707/5 - Castings, Ductile Iron and Austenitic Ductile Iron
MIL-N-25027 - Nut, Self-Locking, 250°F, 450°F, and 800°F.
MIL-H-46855 - Human Engineering Requirements for Military Systems, Equipment and Facilities.
MIL-R-83248 - Rubber, Fluorocarbon Elastomer, High Temperature, Fluid, and Compression Set Resistant

STANDARDS

FEDERAL

FED-STD-H28/2 - Screw-Thread Standards for Federal Services-Section 2, Unified Inch Screw Threads-UN and UNR Thread Forms.

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FED-STD-66 - Steel: Chemical Composition and Hardenability.

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MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-278 - Welding and Casting Standard.
MIL-STD-438 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Submarine Service.
MIL-STD-454 - Electronic Equipment, Standard General Requirements for.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-470 - Maintainability Program for Systems & Equipment.
MIL-STD-471 - Maintainability Verification/Demonstration/ Evaluation.
MIL-STD-721 - Definitions of Terms for Reliability and Maintainability.
MIL-STD-740-1 - Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-740-2 - Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Naval Surface Ships.
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities

HANDBOOK

MILITARY

MIL-HDBK-267 - Guide for Selection of Lubricants and Hydraulic Fluids for Use in Shipboard Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS

MILITARY

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

803-1385850 - Piping Instrument Pressure for All Services.

810-1385917 - Temperature Indicator and Thermowell Selection.

(Application for copies should be addressed to the Portsmouth Naval Shipyard, Code 202.2, Portsmouth, NH 03804-5000).

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 26 - Standard Specification for Aluminum Alloy Sand Castings

A 36 - Standard Specification for Structural Steel. (DoD adopted)

A 108 - Standard Specification for Steel Bars, Carbon, Cold-Finished, Standard Quality. (DoD adopted)

A 148 - Steel Castings, High Strength for Structural Purposes

A 167 - Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip. (DoD adopted)

A 240 - Standard Specification for Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels. (DoD adopted)

A 322 - Steel Bars, Alloy, Standard Grades

A 536 - Ductile Iron Castings

D 2276 - Standard Test Method for Particulate Contaminant in Aviation Turbine Fuels. (DoD adopted)

D 2709 - Standard Test Method for Water and Sediment in Distillate Fuels by Centrifuge. (DoD adopted)

D 3240 - Standard Test Method for Undissolved Water in Aviation Turbine Fuels (DoD adopted)

D 3921 - Oil and Grease and Petroleum Hydrocarbons in Water (DoD adopted)

D 4727 - Standard Specification for Corrugated and Solid Fiberboard Sheetstock (Container Grade and Cut Shapes)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)
J 726 - Air Cleaner Test Code, Recommended Practice. (DoD adopted)

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

STEEL STRUCTURES PAINTING COUNCIL (SSPC)
SP 10 - Near-White Blast Cleaning.

(Application for copies should be addressed to the Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, PA 15213.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 Materials. Purifier materials shall be as specified in table II. Proposed alternate materials equal to or better than the materials specified shall be subject to review and acceptance by NAVSEA. Parts for which specific materials are not specified shall be made of materials similar to those cited for the service intended. Use of dissimilar materials shall be avoided.

TABLE II. Purifier materials.

Application	Material	Specification
Frames	Nodular iron	MIL-C-24707/5
	Steel plate	ASTM A 36
	Cast steel	MIL-C-24707/1, ASTM A 148 (No radiography or magnetic particle inspection)
	Cast aluminum	ASTM B 26
Covers in contact with process fluids	Corrosion resisting steel	ASTM A 167, ASTM A 240, QQ-S-763, class 316, 304
	Corrosion resisting steel forging	DOD-F-24669/6

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TABLE II. Purifier materials - Cont'd.

Application	Material	Specification
Bowl parts	Corrosion resisting steel forging and casting	DOD-F-24669/6 QQ-S-763 grade 316/329
	Corrosion resisting steel	ASTM A 167, ASTM A 240, QQ-S-763, Class 317
Spindle	Forged steel, free machining	MIL-S-24093, class H
	Steels	ASTM A 108, MIL-S-5000, ASTM A 322
Spindle housing	Nodular iron	MIL-C-24707/5, ASTM A 536
	Cast steel	MIL-C-24707/1
Inlet and outlet piping	Corrosion resisting steel	MIL-P-24691/3
O-rings	Fluorocarbon rubber	MIL-R-83248

3.2.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2.2 Prohibited material. The following materials shall be prohibited for use in the manufacture of purifiers covered by this specification:

(a) Magnesium and its alloys.
(b) Cast iron (except nodular iron).
(c) Semi-steel.
(d) Cadmium plating.
(e) Copper alloys in contact with fuel, lubricating oil, or JP-5.
(f) Asbestos.

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3.2.3 Mercury. The purifier shall be free of mercury contamination. During the manufacturing process, tests, and examination, purifiers shall not come in direct contact with mercury or any of its components, or with any mercury containing device employing a single boundary of containment.

3.3 Construction. Purifiers shall separate contaminants from the specified fluid and discharge purified fluid under pressure through closed piping to the applicable ship's system. Water separated from the specified fluid shall be continuously discharged to ship's oily water drain collection system. Separated solids shall collect within the bowl until they are ejected during the shoot cycle, to the ship's oily water drain collection system. The purifier shall utilize water supplied from the ship's potable water system, or dedicated head tank, or both, for self cleaning and other purifier operating functions.

3.3.1 General. Purifiers shall be automatic or semi-automatic as required (see 6.2 and 6.3).

3.3.1.1 Automatic purifiers. Automatic purifiers shall have the following characteristics:

(a) Manual pushbutton motor start on control panel.
(b) Automatic prime when unit is "up to speed" and subsequent to shoot cycle (removal of solids during normal rotation).
(c) Automatic flow of oil after prime.
(d) Automatic control for cyclic or other non-timed initiation of shoot cycle.
(e) Automatic repositioning of bowl operating controls after shoot cycle.
(f) Manual control of purified oil backpressure.
(g) Provision for manual override of automatic features.
(h) Electric motor drive through belts or gears.
(i) Tachometer or counter for measurement of bowl speed.
(j) Attached feed pump (size SC-1 only).
(k) Full discharge bowl (size SC-1 only) or partial discharge bowl (size SC-1 and SC-3).

3.3.1.2 Semi-automatic purifiers. Semi-automatic purifiers shall have the following characteristics:

(a) Manual pushbutton motor start on control panel.
(b) Manual pushbutton or spring return selector switch control on control panel of priming water to bowl.
(c) Manual pushbutton or spring return selector switch control on control panel for flow of oil after prime.
(d) Manual pushbutton or spring return selector switch control on control panel for initiation of shoot cycle.
(e) Manual pushbutton or spring return selector switch control on control panel for repositioning of bowl operating controls and reprime of the bowl after shoot cycle.
(f) Manual control of purifier oil backpressure.
(g) Provision for manual override of electric control features.
(h) Electric motor drive through belts or gears.

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(i) Speed indicator or counter for measurement of bowl speed.
(j) Attached feed pump (size SC-1 only).
(k) Full discharge bowl (size SC-1 only) or partial discharge bowl (size SC-1 and SC-3).

3.3.1.3 Dimensions. Purifiers shall not exceed the dimensions shown in table III.

TABLE III. Limiting dimension.

Dimension	Type	
	SC-1 Meters (feet)	SC-3 Meters (feet)
Length Width Height	1.22 (4) 0.91 (3) 1.52 (5)	1.83 (6) 1.22 (4) 2.13 (7)

3.3.2 Mounts. Purifiers shall be vertical and hard-mounted on a base plate. Thus mounted, purifiers shall be furnished complete and ready for mounting directly to the deck of the ship without the use of a pedestal, unless the use of resilient mounts is cited in any specific acquisition resulting from this specification (see 3.3.2.1).

3.3.2.1 Resilient mounts. When required (see 6.2), resilient mounts shall be in accordance with MIL-M-17508.

3.3.3 Frames. Purifier frames shall be sufficiently rigid to permit lifting, handling, shipment, and installation of the units on board ship without disturbing the alignment of the assembled units, and such that the normal distortion, weaving or vibration of the supporting structures on board ships shall not cause misalignment of rotating elements.

3.3.3.1 Special handling provisions. Special tools or fittings shall be provided by the contractor for lifting items weighing in excess of 16 kilograms (35 pounds) (see 3.9).

3.3.4 Brakes. Size SC-3 purifiers shall be equipped with brakes.

3.3.5 Bearings. Bearing installation and lubrication shall be in accordance with FF-B-171 for surface ships, MIL-B-17931 for submarines and low noise surface ship applications, or FF-B-185, as applicable. In the selection of bearings, consideration shall be given to the fact that rolling and pitching of a ship at sea may introduce unusual radial and axial thrust loads.

3.3.5.1 Bowl spindle bearings. Purifier bowl spindle bearings shall be ball bearings. Other bearings shall be of the sliding contact or rolling contact types. Double row bearings and bearing sets are prohibited unless they are of the self-aligning type.

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3.3.6 Lubricants. Lubricants used for purifier operation, assembly and maintenance shall be selected from those standard lubricants specified in MIL-HDBK-267.

3.3.7 Piping and valves. Connection to ship's piping shall be flanged. Flange size, rating, specification, and surface finish for piping connections shall be as specified (see 6.2). Valves, flanges, and bolting interfacing with ships systems shall conform to MIL-STD-777 for surface ship applications and MIL-STD-438 for submarine applications as specified (see 6.2).

3.3.8 Operating water connections. Connections shall be provided on purifiers for purifier bowl prime, close and shoot water. Quantities for prime, close and shoot water shall be designated by the contractor. The configuration shall minimize quantity requirements. In-line filters required for water purification shall be furnished by the contractor.

3.3.9 Process fluid and water interface control. Control of the interface between liquid seal (prime water) and process fluid shall be achieved by means of a back pressure regulating valve and gravity discs (see 3.3.10). Under all operating conditions back pressure shall be sufficient to prohibit aeration and foaming of purified fuel.

3.3.10 Adjustment for specific gravity of oil. The purifier shall include a set of gravity discs to adjust for changes in specific gravity to ensure achieving the highest possible degree of purification and clarification within the range specified (see 6.2).

3.3.11 Drain connections. Connections shall be provided for casing, water, and sludge drains.

3.3.12 Low effluent switch. Purifiers shall be equipped with a low effluent switch assembly located in the purified process fluid (effluent) discharge line to actuate an alarm by means of a time delay relay. The flow switch shall be in accordance with MIL-S-16032.

3.3.13 Sight glasses. Sight glasses shall be provided in the heavy phase (water) discharge.

3.3.14 Controls and actuators. Sequence control and valve actuators shall be electrically actuated; however, provision for manual operation shall be included. Consideration will be given to the use of other means for actuation, and shall be subject to the approval of the contracting activity.

3.3.15 Overload relay. The overload relay shall be effective for all periods other than the starting period. It shall be the compensated thermal type. An embedded detector type shall not be used.

3.3.16 Vibration safety interlock. Purifiers shall be equipped with a vibration safety interlock to sense excessive vibration during operation and to secure power to the purifier with a time-delayed automatic shutdown.

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3.3.17 Pressure gauges, gauge piping and connections. Pressure gauges, when supplied, shall be 9 centimeters (3.5 inches) in diameter and shall be in accordance with MIL-G-18997, dial size and case design symbol 3. Gauge piping and connections shall be in accordance with Drawing 803-1385850.

3.3.18 Thermometers and thermometer wells. Thermometer wells and bulbs, when supplied, shall be in accordance with Drawing 810-1385917. Thermometers shall be 7.6 centimeters (3 inches) in diameter, in accordance with MIL-I-17244. Use of mercury thermometers is prohibited.

3.3.19 Oil dipstick and sight glass. Purifiers shall be provided with an oil dipstick or sight glass to determine the lube oil level within the drive mechanism case.

3.3.20 Thread fit. Screw threads shall be in accordance with FED-STD-H28/2.

3.3.20.1 Alignment critical fit. Where alignment must be maintained, fitted bolting or positive interference methods shall be used. Fit shall be:

Nominal size centimeters (inches)	Maximum clearance (plus) and interference (minus), centimeters
1.27 to 2.86	plus 0.0013 (minus 0.0005)
(1/2 to 1-1/8)	plus 0.0025 (minus 0.0010)

3.3.21 Fastener application. Except where torque-controlled, prestressed bolts and nuts are used, not less than one thread but not greater than four threads shall protrude beyond the crown of the nut. With plastic insert self-locking nuts, the end of the thread runout shall be not less than one thread above the top of the plastic insert. Washers shall not be used under the nut for the purpose of reducing thread protrusion. Thread engagement for the setting end of a stud shall be such that the shear load strength of the engaged threads shall be greater than the tensile load strength of the stud.

3.3.21.1 Thread engagement for materials of similar and dissimilar tensile strengths. For materials having similar mechanical properties, the full thread engagement shall be not less than one major diameter (1D). For materials having differing tensile strength, the minimum engagement of stud setting threads shall be computed in accordance with FED-STD-H28/2, using the maximum expected tensile strength of the stud material and minimum specified (or expected, if no minimum is specified) tensile strength of the body material, plus one thread, but in no case less than 1D.

3.3.21.2 Through tapping and bolting. Through tapping is permissible only if metal thickness is sufficient for 1D full thread engagement plus thread run-out and end bevel, and shall be used only where necessary to maintain watertight or structural integrity. Bottom-tapped holes shall have full threads for the entire depth. Through bolting shall be used wherever possible. If use of such bolting is not possible, studs, tap bolts, or machine screws shall be used. Thread engagement in blind holes shall be not less than 1D.

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3.3.21.3 Foundation bolts. Foundation bolts or studs shall be not less than 1/2 inch in diameter.

3.3.21.4 Self-locking nuts. When required to prevent loosening due to shock or vibration, self-locking nuts in accordance with MIL-N-25027 shall be used. Self-locking nuts having plastic inserts or self-locking nuts of the all-metal, distorted-top type shall be used at temperatures which shall be not greater than 121 degrees Celsius (°C) (250 degrees Fahrenheit (°F)). The following documents apply to nuts of the indicated material:

Nickel-copper alloy	- MS17828.
Carbon steel	- MS17829; modified to specify MIL-S-1222, grade 2H for use with alloy steel bolts, or grade 2 for use with carbon steel bolts.
Corrosion resisting steel	- MS17830; modified to specify 15303 of FED-STD-66.

3.3.22 Pipe plugs. Purifier pipe plugs shall not have tapered threads. Pipe plugs shall be straight machine threaded. Tight sealing shall be ensured by use of O-rings.

3.3.23 Doweling. Each component part of an assembled unit supported directly by a frame or bedplate shall be doweled to facilitate reassembly and maintenance of alignment. Dowels shall be unequally spaced or be of different diameters to avoid incorrect assembly.

3.3.24 Machined surfaces. Bearing and seating surfaces of frames or bedplates shall be finish machined.

3.3.25 Welding. Welding shall be in accordance with MIL-STD-278. Loose welding beads, slag, and spatter shall be removed.

3.3.26 Painting. Non-working surfaces, normally painted, shall be finished as specified in the following procedures:

(a) Surface preparation prior to painting in accordance with SSPC surface preparation SP-10.
(b) Two coats epoxy paint formula 151 and one coat of formula 150 in accordance with MIL-P-24441, MIL-P-24441/1, and MIL-P-24441/2, average 3 mils dry film thickness per coat.

3.3.27 Purifier pumps. An attached feed pump is required for size SC-1 purifiers. Purifiers shall have attached discharge pumps or shall be equipped with paring devices (or similar devices) for discharging process fluid under pressure to the ship's piping. Unless otherwise specified (see 6.2), no other pumps shall be furnished. Attached pumps, when supplied, shall be type III, class N-4 or O-3 in accordance with MIL-P-19131. If provided, the capacity of each discharge pump shall be approximately 110 percent of the rated capacity of the corresponding feed pump. Pumps shall be furnished with built-in spring loaded bypass valves. A thermometer shall be installed between the discharge of the feed pump and the inlet to the purifier bowl. Unless otherwise specified (see 6.2), the pumps and the purifier shall be driven by a common motor.

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3.3.28 Oil heaters. When specified (see 6.2), process fluid heaters shall be supplied with the purifiers. Heaters shall heat process fluid from the temperature specified (see 6.2) to optimum purification temperature as specified in 3.5.1, with the process fluid flowing through the heater at the rated capacity of the feed pump.

3.3.28.1 Heater type. Heaters shall be in accordance with MIL-H-17428 (steam type), or MIL-H-24299, type B (electric type) (see 6.2). Fuel steam heaters shall be in accordance with MIL-H-16313.

3.3.29 Motors. When required (see 6.2), alternating current (ac) motors shall conform to MIL-M-17060 and shall have the following characteristics:

Horsepower	See 3.3.29.1
Service	A
Speed classification	Constant
Duty	Continuous
Enclosure	Totally enclosed or dripproof
Bearings	In accordance with FF-B-171; MIL-B-17931, if low noise application
Ambient temperature	50°C (122°F)
Insulation	B or F, with sealed insulation system, or class H
Type	Squirrel cage induction
Voltage rating	440 volts alternating current (Vac), three-phase, unless otherwise specified (see 6.2)
Acceleration allowance	As required (see 3.3.31)
Low noise application	As required (see 6.2)
Mechanical balance	Precision; superprecision if low noise application

If direct current (dc) motors are required (see 6.2), MIL-M-17413 shall apply.

3.3.29.1 Horsepower (HP) rating. The horsepower rating of each motor shall be not less than the sum of the maximum brake horsepower (bhp) of the driven components under any condition of service. The actual motor rating shall be in accordance with horsepower sizes specified in MIL-M-17060 or MIL-M-17413, as required. If the maximum bhp of the equipment is less than 2, and if there is a possibility of an unpredictably high frictional load due to improper adjustment of glands, or to some other cause, then the maximum bhp shall be multiplied by a safety factor of 1.5 - Maximum bhp before selecting the next larger Navy standard rating. 4

3.3.29.2 Thermal protection. Thermal protection shall be provided for the purifier motor stator and motor bearings in accordance with MIL-M-17060, with the provision for de-energizing the motor control. The thermal protection shall include motor-mounted monitor control with a plug-in relay having a minimum of one pair of normally open and one pair of normally closed contacts. The temperature sensing system shall include a minimum of three thermal sensors imbedded in the stator windings and one thermal sensor for each motor bearing. Positive temperature coefficient (ptc) thermal sensors shall be used and shall have a temperature

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coefficient of resistance at the switchpoint of not less than 15 percent per degree Celsius.

3.3.30 Controllers. When specified (see 6.2), motor controllers shall be provided. Controllers shall conform to MIL-C-2212 and shall have the following characteristics:

Enclosure	Dripproof
Operation	Magnetic
Type	Across line, except type SC-3 purifier with direct drive shall be wye-delta
Function	Motor starting and purifier control
Duty	Continuous
Protection	Overload - under voltage (LVP)
Ambient temperature	50°C (122°F)
Voltage rating	440 Vac, 3-phase, unless otherwise specified (see 6.2) (dc voltage requirements, if applicable, shall be as specified (see 6.2))
Performance	Automatic or semi-automatic (see 6.2)
Master switches	Local, manual
Abnormal vibration frequencies	When not covered by MIL-C-2212, frequency shall be as specified (see 6.2)
Low noise application	When required (see 6.2), insulation shall be class B, with class A for coils only.

3.3.31 Acceleration allowance. Motors and controllers shall permit acceleration of the unloaded purifier to full operation speed in minimal time, as indicated by the purifier manufacturer, without exceeding permissible temperature rise in any components or causing wear due to slippage of drive.

3.3.32 Interchangeability. All identically identified components shall be functionally and physically interchangeable, without degradation of performance, reliability, or operating characteristics and without selective assembly or modification except for calibration and adjustment. Repair parts shall be interchangeable with, and identified identically to, the part they replace. Notwithstanding the above, parts balanced as complete assemblies shall not be exchanged without rebalancing the complete assembly per manufacturers specifications.

3.3.33 Accessibility. Maximum accessibility shall be provided (within space limitations) to parts which require scheduled maintenance and to affect repairs. The configuration shall be such that the purifier can be opened without breaking any piping connections. Removal of major operational parts shall not be necessary to permit accessibility for maintenance, inspection, and repair. The human engineering considerations specified in MIL-STD-1472 and MIL-H-46855 shall not be degraded during detail design and construction.

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3.4 Performance.

3.4.1 Capacity. Purifier sizes and capacities shall be as shown in table I for the fuels and lube oils as specified (see 6.2). The capacity of a purifier shall be the flow rate maintained for a period of 1 hour when supplied with process fluid specified in table I during which time contaminants in purified process fluid shall not exceed the contaminant level specified in 3.4.2. Tests for establishing capacities are specified in 4.6.4.

3.4.1.1 Emergency capacity. The purifier shall accept influent of 100 percent water for 5 minutes without any water being discharged from the clean process fluid outlet. The water discharge capacity shall be equal to the purifier inlet flow.

3.4.1.2 Surplus capacity. Purifiers without attached feed pumps shall purify fuel in excess of at least 8 percent of its rated capacity. (This requirement is necessary to accommodate shipboard transfer pump flow rates that exceed their rated capacity and provides the design activity a limited degree of flexibility when sizing the fuel transfer pump).

3.4.1.3 Alternate fuels. The purifier shall purify JP-5 in accordance with MIL-T-5624.

3.4.2 Effluent quality. The contaminant level in purified process fluid shall not exceed the level specified in table IV when operating at the capacities specified in 3.4.1 (see 4.6.8).

TABLE IV. Maximum allowable contaminants.

Fluid	Contaminant	
	Water	Solids
Lubricating oils MIL-L-9000 MIL-L-17331 Fuel MIL-F-16884	N/A 0.05 percent by volume 30 parts per million (p/m)	0.02 percent by volume 0.02 percent by volume 4 milligrams per liter or 10 percent of influent solids

3.4.3 Solids retention capacity. The solids retention volume of the purifier shall be not less than specified in table V.

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TABLE V. Solids retention capacity.

Size	Effective solids 1/ retention volume (gallons)
SC-1	0.1
SC-3	1.5

1/ No degradation of effluent quality.

3.4.4 Shock. The purifier shall meet the requirements for grade A, type A, class I or II (see 6.2) high impact shock in accordance with MIL-S-901. The purifier shall restart and resume normal operation if it shuts down when subjected to the test requirements specified in MIL-S-901 for grade A, class I, type A. Equipment shall be hard-mounted to a base plate for test purposes. The determination of stresses resulting from the shock test shall be accomplished by the application of analytical mechanics. The allowable stress in components due to shock shall be the yield strength (0.2 percent offset of the material). Normal stresses in parts, such as the tensile stress in a bolt due to initial pull-up, shall not be added to the shock stresses. The maximum accelerations possible under the shock tests shall be utilized in the design of hold-down bolts, supporting feet or lugs, and main structural member of the equipment. Failure or permanent deformation shall not occur as a result of shock. Components of an assembly mounted on a common rigid base need not be designed for higher accelerations than the values for which the whole assembly is designed unless the contractor has reason to believe that higher accelerations will be experienced by the components.

3.4.5 Vibration.

3.4.5.1 Mechanical. The equipment shall conform to MIL-STD-167-1, type II.

3.4.5.2 Environmental. The equipment shall withstand the environmental vibration requirements specified in MIL-STD-167-1, type I.

3.4.6 Noise. The equipment shall be in accordance with the acceptance criteria specified in MIL-STD-740-1 (airborne) and MIL-STD-740-2 (structureborne), when tested (see 4.6.11).

3.4.7 Electromagnetic interference. The equipment shall conform to the electromagnetic interference suppression requirements of parts 1 and 5 of MIL-STD-461 for surface ships (class A4). The acceptance testing shall be in accordance with MIL-STD-462.

3.5 Environmental conditions. Equipment shall operate continuously at rated capacity under any combination of the range of environmental conditions specified herein.

3.5.1 Temperatures. The purifier shall operate within the temperatures specified in table VI.

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TABLE VI. Temperatures.

	F-76 fuel °C (°F)	Turbine lube oil °C (°F)	Diesel engine lube oil °C (°F)
Influent temperature range	16-27 (60-80)	66-77 (150-170)	74-85 (165-185)
Ambient temperature range	4-50 (40-122)	4-50 (40-122)	4-50 (40-122)

3.5.2 Humidity. Purifiers shall operate and meet all specified requirements when subjected to relative humidity of 0 to 100 percent.

3.5.3 Pitch and inclined operation. The equipment shall suffer no damage and shall operate to the requirements of this specification, unless otherwise specified, when subjected to the motions specified in table VII.

TABLE VII. Motions. 3/ 4/ 5/

Motion attitude	Surface ships		Submarines	
	radians	(degrees)	radians	(degrees)
Pitch (from normal either direction) 1/	0.17	(10)	0.175	(10)
Trimmed down (from normal horizontal)	0.087	(5)	0.524	(30)
Roll (from vertical) 2/	0.785	(45)	1.047 0.524	(60) surfaced (30) submerged
Permanently listed (either side or vertical)	0.262	(15)	0.262	(15)

1/ Time of cycle, 10 seconds.
2/ Time of cycle, 15 seconds.
3/ The purifier shall maintain lubrication, without loss of oil to the drive assembly. The purifier shall continue to operate, without the need for re-priming, after recovering from the subject motion attitude.
4/ Trim and list conditions shall be considered to occur simultaneously.
5/ Roll and pitch conditions shall be considered to occur simultaneously.

3.6 Reliability and maintainability. Definitions of terms for reliability and maintainability shall be in accordance with MIL-STD-721, MIL-STD-785, and MIL-STD-470.

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3.6.1 Reliability. Mean time between failure (MTBF) for the purifier shall be not less than 1500 hours (see 6.3).

3.6.2 Maintainability. Mean time to repair (MTTR) for the purifier shall not exceed 2 hours (see 6.3).

3.6.3 Failure. For purposes of reliability calculation, the purifier is in failed state if it cannot be started, or if a malfunction occurs that forces the purifier to be shut down prior to the completion of its scheduled test time or operating time. In addition, the purifier shall be considered to fail if the capacity (see 3.4.1) or the quality of purification (see 3.4.2) is not in accordance with the requirements specified herein, even if the unit has not experienced a forced shutdown. Failures shall be classified as either relevant or nonrelevant. Only relevant failures shall be chargeable for verifying reliability requirements.

(a) Relevant failures are caused by:
  (1) Equipment design defects.
  (2) Equipment manufacturing defects.
  (3) Parts defects.
  (4) Unknown.
  (5) All other causes not specifically listed as non-relevant.

(b) Non-relevant failures are caused by:
  (1) Accident, mishandling, or improper storage.
  (2) Operator or procedural error.
  (3) External test equipment or facility failures.
  (4) Drawing, maintenance manual, or other documentation errors, provided prompt correction is made and the correction is shown to eliminate future similar failures (this shall not be construed to include design errors).
  (5) Failures of multiple, simultaneous or immediately sequential nature (only the initial failure shall be counted in such cases).
  (6) Direct result of relevant or non-relevant failures of another item of equipment (secondary failures).
  (7) Improper maintenance induced failures.
  (8) Foreign object damage, unless caused by failure of protective device built into contractor's equipment.

Failure reporting shall include all failures, whether non-relevant or relevant.

3.7 Identification plates. Identification plates shall be furnished on each purifier and shall be type A or B in accordance with MIL-P-15024 and MIL-P-15024/5, except that plastic identification plates are not acceptable. Identification plates shall be secured to equipment with corrosion-resistant metallic screws.

3.7.1 Identification data. Purifier identification plates shall contain the following data:

(a) Manufacturer's name.

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(b) "Self Cleaning Centrifugal Purifier" (followed by flow rate and service application).
(c) Manufacturer's model.
(d) Navy size designation.
(e) Date of manufacture (month and year).
(f) Manufacturer's serial number.
(g) Contract number (and item number for multiple unit orders).
(h) National stock number.
(i) Nominal bowl speed.

3.7.2 Other identification plates. Each motor and each accessory unit shall have an identification plate in accordance with the applicable equipment specification.

3.8 Safety criteria. Equipment shall ensure safety to operating and maintenance personnel (see 6.3). Purifiers shall have their bowls enclosed to prevent free vapor from escaping. External moving parts, which are a potential hazard to personnel, shall be avoided wherever practicable. When their use is unavoidable, positive protection in the form of a guard shall be provided. Sharp corners and projections, which may cause injury to personnel or on which clothing may catch, shall be avoided. The following are some of the known safety criteria that apply to auxiliary machinery; others are specified in MIL-STD-454 and MIL-E-917.

(a) Exposed rotating or reciprocating components, such as couplings, linkage, belts, chains, and flywheels, shall be enclosed with a safety guard to prevent accidental personnel contact with the moving part.
(b) When an equipment specification requires the presence of a hazardous material as a component of shipboard equipment, conspicuous indication on the equipment of the presence of such material, together with necessary warnings and instructions, is required.

3.9 Special tools. Special tools shall be furnished in the quantity specified (see 6.2 and 6.8). Special tools are defined as those items required for shipboard maintenance, servicing, and repair not listed in the Federal Supply Catalog. (Copies of this catalog may be consulted in the office of the Defense Contract Administration Services Management Area (DCASMA)).

3.10 Workmanship. The purifier shall be uniform in quality and shall be free from irregularities, defects, or foreign matter that could adversely affect safety, performance, reliability, or durability. Parts of the purifier, before and after painting, shall be clean and free of sand, rust, dirt, pits, sprues, scale, and other harmful extraneous material which might detract from the intended operation, function, or appearance of the equipment. Fasteners shall be fully engaged and bolt heads and nuts shall be flat on washers or mating surface.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as

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otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program (see 6.3). The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be performed at a laboratory acceptable to the Government on a sample purifier produced with equipment and procedures used in normal production.

4.3.1 Sampling for first article inspection. One sample unit of each size (see table I) to be supplied shall be subjected to first article inspection. The sample purifier shall be submitted for first article tests with pumps attached and driven by the same motor that drives the purifier.

4.3.2 Inspection routine. Sample units shall be subjected to the examinations specified in 4.5 through 4.5.3 and the tests specified in 4.6 through 4.6.12 (see 6.3).

4.3.3 Failure. Failure to pass the examinations (see 4.5), or failures other than those allowed during testing (see 3.6.3) shall be cause for failure of first article testing.

4.4 Quality conformance inspection. Each purifier shall be examined and tested as specified in 4.5 through 4.6.2 (see 6.3).

4.5 Examinations. The manufacturer shall examine every completed purifier in accordance with 4.5.1 through 4.5.3. If one or more defects are found in any purifier, the unit shall be rejected. The manufacturer shall maintain for a period of three years after contract completion all records of inspections, tests, or resulting rejections.

4.5.1 Fabrication. Welding shall be examined to ensure conformance to 3.3.25.

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4.5.2 Painting. Every purifier shall be examined at each step of the painting to ensure conformance to 3.3.26.

4.5.3 Workmanship. Every purifier shall be examined to ensure conformance to 3.10.

4.6 Tests. Purifiers shall be tested as specified in 4.6.1 through 4.6.12.

4.6.1 Spin-test. Prior to installation in the purifier, the bowl assembly shall be spin-tested at 125 percent of the maximum design operation speed for a period of at least 15 minutes.

4.6.2 Driver test. The purifier, with driver, shall be operated at rated speed and full capacity for a period of not less than 1 hour to determine smoothness of running, satisfactory operation of controls, and proper assembly. Quantitative measurements of purification are not required. The motor controller used during this test shall be identical to the one to be supplied with the purifier. The type of fluid circulated through the purifier during the test shall be the fluid specified for the particular size and capacity (see 3.4.1).

4.6.3 Capacity test. The sample unit chosen for first article testing shall be tested for capacity to ensure conformance to 3.4.1. For the tests specified in 4.6.3.1 and 4.6.3.2, effluent quality tests, test conditions, acceptable performance, and test methods shall be performed in accordance with section 4.6.7.

4.6.3.1 Capacity test for service fluid. The purifier shall be operated for a period of not less than 1 hour to verify capacity. The service fluid used shall be as specified in 6.2 (b).

4.6.3.1.1 Surge load test. A five minute surge load test, using 100 percent influent water, shall be performed (see 4.6.4.1) at the flow specified in table I.

4.6.3.2 Surplus capacity test. The purifier shall be operated using F-76 fuel, as specified in MIL-F-16884, for a period of not less than one hour. The feed rate shall be not less than 20 percent in excess of the rated capacity (i.e., 135 gpm for size SC-3 purifier).

4.6.4 Reliability test. The sample unit chosen for first article testing shall be operationally tested for not less than 1800 hours to ensure conformance to 3.6 (see 6.3). Each operational period shall be not less than 7 hours. For size SC-1, the operating fluid used during the test shall be turbine lube oil in accordance with MIL-L-17331. For size SC-3, the operating fluid shall be fuel in accordance with MIL-F-16884. Reliability test may be conducted concurrently with the tests specified in 4.6.3 and 4.6.6 through 4.6.8. An effluent quality test shall be performed on the first article sample unit at 10, 600, 1200 and 2000 hours of operation. One 5-minute surge load test shall be performed on the sample unit at every 300 hours of operation for 1800 hours.

4.6.4.1 Surge load test. The sample unit shall process influent with water concentrations of 25, 50, and 100 percent, without discharging any water from the clean process fluid effluent.

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4.6.5 Maintainability test. The contractor shall demonstrate that the equipment design meets the quantitative requirements of MIL-STD-471, test method 4 (see 6.3).

4.6.6 Inclination test. For 24 hours of the reliability test, the purifier shall be at an inclination of 0.262 radians (15 degrees).

4.6.7 Effluent quality tests. Test conditions, acceptable performance, and test methods of determining acceptable performance shall be as specified in 4.6.7.1, 4.6.7.2, and 4.6.7.3, respectively.

4.6.7.1 Test conditions. Test conditions shall be as follows:

(a) Flow rate. The flow rate shall be as specified in table I, plus 10 percent, minus 0 percent (or as specified for surplus capacity test of 4.6.3.2).

(b) Water in influent. The amount of water in influent shall be 1.0 ± 0.1 percent of the fuel or lube oil flow. Water shall not be injected to diesel engine lube oil MIL-L-9000.

(c) Solids contaminants. Solids contaminant shall be air cleaner fine test dust in accordance with SAE J 726, in the quantities specified as follows:

Fuel:
(1) 100 +/- 20 milligrams per liter.
(2) 700 +/- 300 milligrams per liter.

Lube oil:
(3) 2.0 +/- 0.7 grams per liter.

(d) Temperature. For fuel: 10 ± 5.5 °C (50 ± 10 °F. For turbine lube oil (MIL-L-17331): 74 ± 4.4 °C (175 ± 10 °F. For diesel engine lube oil (MIL-L-9000): 74 ± 5.5 °C (160 ± 10 °F).

4.6.7.2 Acceptable performance. Acceptable performance shall be as follows:

(a) Solids in effluent. The purified process fluid effluent shall have either:

Fuel:
(1) An average weight of solids not greater than 4.0 milligrams per liter (influent solids specified in 4.6.7.1 (c)(1)).

(2) An average weight of solids not greater than 10 percent of the influent solids (influent solids specified in 4.6.7.1 (c)(2)).

Lube oil:
(3) An average solids content of 0.02 percent by volume.

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(b) Oil or fuel in heavy phase discharge (water). The oil or fuel content in water effluent shall be not greater than 500 parts per million for any sample.

(c) Free water in purified effluent. The average free water content in the purified effluent shall be as follows:

(1) Fuel - less than 30 parts per million or less.
(2) Turbine lube oil - 0.05 percent by volume or less.

4.6.7.3 Test methods. The influent and effluent samples shall be taken simultaneously. Alternate or modified test procedures (i.e., sampling and analysis) shall be subject to review and acceptance by the contracting activity.

4.6.7.3.1 Influent test methods. Influent quality test methods shall be performed to verify conformance to 4.6.8.1 (b) and (c) as follows:

(a) Sampling procedures. The sample cock shall be fully opened and 200 milliliters shall be drawn. The 200 milliliters shall be discarded. Samples shall then be obtained as follows:

(1) Undissolved (free) water. At least six influent samples shall be drawn for free water determination.

(2) Solids contamination. At least six influent samples shall be drawn into a 32 ounce bottle for solids content determination.

(b) Undissolved (free) water determination. Undissolved water shall be determined in accordance with ASTM D 2709 (BS&W). Undissolved water shall be determined to nearest p/m. The water in oil shall be as specified in 4.6.8.1 (b) (not required for engine lube oil).

(c) Solids content determination. The inorganic solids content shall be determined in accordance with ASTM D 2276. Analysis shall be performed on entire sample. The solids contaminant injection shall be as specified in 4.6.8.1 (c).

4.6.7.3.2 Effluent test methods. Effluent quality test methods shall be performed to verify conformance to 4.6.7.2 (a) and 4.6.7.2 (c) as follows:

(a) Sampling procedures. The sample cock shall be fully opened and 200 ml shall be drawn. The 200 ml shall be discarded. Samples shall then be obtained as follows:

(1) Undissolved (free) water. At least six effluent samples shall be taken for free water determination (not required for engine lube oil).

(2) Solids contamination. At least six effluent samples shall be taken for solids content determination.

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(b) Solids content determination. The inorganic solids content in fuel shall be determined in accordance with ASTM D 2276 (Gravity Analysis). The solids content in lube oil shall be determined in accordance with ASTM D 2709 (BS&W).

(c) Undissolved water in fuel. The undissolved water in aviation fuel shall be determined in accordance with ASTM D 3240. When this method is used for F-76, 5 p/m result represents 40 p/m free water content in the F-76 fuel sample. The undissolved water in oil shall be determined in accordance with ASTM D 2709 (BS&W).

4.6.7.3.3 Heavy phase (water) discharge test method. Heavy phase (water) discharge test method shall be performed to verify conformance to 4.6.7.2 (b) as follows:

(a) Oil or fuel in water discharge. At least one sample shall be collected for oil or fuel in water discharge determination.

(b) Oil or fuel in water discharge determination. The oil or fuel in water analysis shall be determined in accordance with ASTM D 3921. Results shall be expressed in p/m by volume using the specific gravity of the given process fluid.

4.6.8 Shoot cycle solids rejection test. The first article sample unit shall operate for 2000 hours under the conditions specified in 4.6.5 without requiring manual cleaning.

4.6.9 Vibration test. The first article sample unit shall be subjected to type I (see 6.2) and type II environmental and mechanical vibration in accordance with MIL-STD-167-1 and shall meet the requirements specified in 3.4.5. The frequency range for type I vibration shall be as specified in the contract or order (see 6.2).

4.6.10 Noise test. The purifier shall be noise tested as follows:

4.6.10.1 Airborne noise test. The sample purifier shall be tested in accordance with the procedures specified in MIL-STD-740-1 and shall meet the performance requirements specified in 3.4.6.

4.6.10.2 Structureborne noise test. When specified, every completely assembled purifier shall be tested in accordance with the procedures specified in MIL-STD-740-2 and shall meet the performance requirements specified in 3.4.6 (see 6.2).

4.6.11 Shock tests. The first article sample unit shall be subjected to the high impact grade A shock test specified in MIL-S-901 and shall meet the requirements specified in 3.4.4 (see 6.3).

4.6.11.1 Resubmission for shock testing. Equipment previously accepted will not be resubmitted for shock test, except when evidence of low shock resistance develops in the units installed. The Command or agency concerned will initiate

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action with the contractor for correction of deficiencies, and may require shock tests at any time.

4.6.11.2 Shock testing with drivers. Unless otherwise specified (see 6.2), the test purifiers shall be shock tested with drivers. Purifiers shock tested with one driver will not be required to be shock tested again when supplied with a different shock qualified driver. Drivers are subject to shock tests in accordance with the applicable equipment specifications.

4.6.11.3 Failure criteria. The purifier shall be considered to have failed the shock tests in the event of any of the following:

(a) Breakage of any parts, including mounting bolts.
(b) Appreciable distortion or dislocation of any part, such as shaft, mounting feet, bearings (except for anti-friction bearing brinelling), or other parts which would render the unit unusable in any emergency.
(c) The purifier cannot be restarted or exhibit acceptable performance as specified in 4.6.5.

4.6.12 Electromagnetic interference. To determine conformance to 3.4.7, the purifier shall be tested as specified in applicable provisions of MIL-STD-462.

4.7 Individual components. Individual components, such as pumps, controllers, heaters, and so forth, shall be examined and tested at the place of manufacture in accordance with the requirements of the applicable component specification.

4.8 Inspection of packaging. Sample packages and the inspection of the packaging (preservation, packing and marking) for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging or preparation for delivery requirements of referenced documents listed in section 2, see 6.7.)

5.1 Packaging requirements. The packaging (preservation, packing and marking) requirements of purifiers shall be in accordance with MIL-P-16789 for level (A, C or commercial) of preservation; level of packing (A, B, C, or commercial), and marking, including packaging acquisitioning options therein as specified (see 6.2). In addition, the following applies:

(a) Navy fire-retardant requirements.

(1) Treated lumber and plywood. When specified (see 6.2), all lumber and plywood, including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

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Levels A and B - Type II - Weather resistant.
Category 1 - General use.

Level C - Type I - non-weather resistant.
Category 1 - General use.

(2) Fiberboard. Fiberboard used in the construction of class domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packing forms shall meet the flame-spread index and the specific optic density requirements of ASTM D 4727.

5.1.1 Marking. In addition to the marking required under 5.1, purifiers shall be marked as follows:

(a) Critical close tolerance equipment. Unit packs and shipping containers and unpacked shipments shall be marked with the following:

"CRITICAL, CLOSE TOLERANCE
OPERATING EQUIPMENT
HANDLE WITH CARE
DO NOT DROP OR SUBJECT
TO SHOCKS OR JARS"

Markings shall be stenciled, red color, and applied on two sides and both ends of the container or shipment, letters shall be minimum 1-1/2 inches high, and shall be of such size as to be legible. In addition, arrows and the word "UP", center of balance, sling or lifting point markings shall apply.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Purifiers covered by this specification are intended for use aboard Naval ships for the removal of water and solids from fuel and lube oils.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number and date of this specification.
(b) Purifier size and operating fluid (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) When first article inspection is required (see 3.1).
(e) Whether purifiers are automatic or semi-automatic (see 3.3.1).
(f) When resilient mounts are required (see 3.3.2.1).
(g) Ship applicability (surface ship or submarine) (see 3.3.7).
(h) Piping connection requirements (see 3.3.7).

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(i) Range of specific gravity adjustments (see 3.3.10):

(1) For F-76 in accordance with MIL-F-16884.
(2) For diesel engine lube oil in accordance with MIL-L-9000, and turbine lube oil in accordance with MIL-L-17331.

(j) When purifier pumps other than those specified are required (see 3.3.27).
(k) When independent pump drive is required (see 3.3.27).
(l) When oil heaters are required (see 3.3.28).
(m) Oil heater type (see 3.3.28.1)
(n) Oil temperature (see 3.3.28).
(o) Motors (see 3.3.29).

(1) Type (ac or dc).
(2) Ac voltage rating, if other than specified.
(3) Low noise application.

(p) Controllers (see 3.3.30):

(1) Ac or dc voltage (if other than specified for ac rating).
(2) Whether automatic or semi-automatic controllers are required.
(3) Abnormal vibration frequency when not covered by MIL-C-2212.
(4) Low noise application.

(q) Noise requirements if other than specified (see 3.4.6).
(r) Special tools (see 3.9).
(s) Frequency range for type I vibration testing in accordance with MIL-STD-167-1 (see 4.6.10).
(t) Level of preservation and packing and other packaging acquisition options required (see 5.1).
(u) When fire-retardant material is required (see 5.1(a)).
(v) Shock requirements if other than specified (see 3.4.4).
(w) If shock tests shall br conducted without drivers (see 4.6.12.2).
(x) When every completed purifier shall undergo the structureborne noise test (see 4.6.11.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference paragraph	DID number	DID title	Suggested tailoring
3.6.1	DI-R-7085	Failure mode, effects, and criticality analysis report	----
3.3.1 and appendix A	DI-DRPR-80651	Engineering drawings	Level 1 Drawing number - Contractor Design activity - Contractor
3.6.1	DI-RELI-80250	Reliability test plan	----
3.6.1	DI-RELI-80251	Reliability test procedures	----
3.6.2	DI-MNTY-80831	Maintainability/ testability demonstration test plan	----
3.8	DI-SAFT-80100	System safety program plan	----
3.8	DI-SAFT-80101	System safety hazard analysis report	----
3.10	DI-V-7002	Provisioning parts list	----
3.10	DI-V-7016	Provisioning and other preprocurement data	----
4.3.2	UDI-T-23732	Procedures, test	----
4.4	DI-NDTI-80809	Test/inspection Reports	----
4.6.5	DI-RELI-80252	Reliability test report	----
4.6.6	DI-MNTY-80832	Maintainability/testability demonstration test report	----
4.6.12	DI-ENVR-80708	Shock test report	----

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract. Technical content should include the requirements of appendix B, titled "Manual Technical Content Requirements."

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested

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as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.6.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 Sub-contracted material and parts. The packaging or preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Special tools. It is desired that one set of special tools as defined in 3.9 and 6.2 be acquired for each ship's compartment in which purifiers are installed. The number of sets required should be established by the acquisition requesting activity.

6.9 Bid data. The following information should be submitted with each bid:

(a) Manufacturer's model and type designation of purifier.
(b) Sectional assembly of purifier or reference to previously approved drawings.
(c) A drawing showing overall dimensions of the unit or reference to previously approved drawings.
(d) Horsepower of driving motors.
(e) Output shaft speed of driving motor and of rotating element.
(f) Weight of complete unit.
(g) Manufacturers of major components such as motors and controllers.
(h) Description of controls.

6.10 Patent notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government acquisition operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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6.11 Reconditioning of shock-tested equipment. Purifiers which have been subjected to high-impact shock tests and have successfully passed this test shall be considered acceptable for such service as the contracting agency concerned may authorize. Such equipment will be returned to the manufacturer for reconditioning as follows:

(a) Minor deformations of mounting feet shall be corrected.
(b) Minor deformations affecting alignment of parts or components shall be corrected.
(c) Each bearing shall be replaced.
(d) Each part shall be carefully examined by the contractor and any part considered substandard shall be replaced.
(e) The rotating elements shall be rebalanced.
(f) The motor shall be corrected as required by the applicable motor specification.
(g) The reconditioned purifier shall be tested in accordance with 4.6.2.

Minor deformations (see items (a) and (b) above) are defined as those which do not cause unqualified rejection of the equipment under high-impact shock test, but which are in excess of the design dimensional tolerances specified on the applicable drawings.

6.12 Subject term (key word) listing.

Actuators
Bearings
Brakes, bowl
Covers
Frames
Gauge, pressure
Housing
Motor
Mounts
Ports, observation
Relay, overload
Sight glasses
Spindle
Thermometer

Preparing activity:
Navy - SH
(Project 4330-N095)

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APPENDIX A

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that should be included on drawings when required by the contract or order. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENT

20.1 Government document.

20.1.1 Specification. The following specification forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

GGG-P-781 - Puller Attachment, Mechanical, and Puller Set, Mechanical.

(Unless otherwise indicated, copies of military specifications are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

30. DRAWING CONTENT

30.1 Drawing content. When required by the contract or order, drawings shall contain the information specified in 30.1.1 through 30.1.7.

30.1.1 Drawings (general). The drawings shall contain the following characteristics:

(a) A sectional assembly drawing. Every part in the assembly shall be shown. Side or partial views may be used to show parts not otherwise shown in the main section view. Clearances, tightening torques and thread lubrication requirements shall be shown on the assembly drawing.

(b) Complete parts list. The parts list shall include an indication of each part required to be furnished as an on board repair part (this is not a repair parts list). The parts shall be identified by the military standard part number when available.

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(c) Flow diagram and interface drawing showing power, pressure, temperature and flow requirements (as applicable) for all connections to ship's systems.
(d) Controller drawing showing all indicators, alarms, and controls. It shall include a logic diagram and a parts list.

30.1.2 Outline drawings. Outline drawings and certification data shall be furnished as supplemental drawings to all sectional assembly and parts list drawings. Separate outline drawings, drawings lists, and certification data shall be furnished under each contract, unless the complete equipments covered by the outline drawings and the referenced drawings are identical in all respects. Outline drawings, in addition to the requirements specified in the contract or order, shall contain the following:

(a) Dimensional outline assembly drawing of the purifier with its motor, bedplate, and attached auxiliaries.
(b) Table of weights of individual components and weight of complete unit (dry, and in the operating condition). This shall include the weight of the subbase, fluids, piping, and any other weight that may contribute to loading of the mountings.
(c) Shipbuilder's connections showing size, type, and dimensions of flanges.
(d) Radii of gyration of complete assembly about each of the three principal axes (required when equipment is sound isolated).
(e) Identification of high-impact shock tests, nondestructive tests, and other tests, if performed, and the Command or agency letters approving these reports or extension of any tests.
(f) In addition, the outline drawing shall include overall dimensions of the complete unit, including locating dimensions of fittings and connections, the space required for removal and replacement of parts for maintenance, and mounting information.
(g) Lifting instructions.

30.1.3 Manufacturer's use only notes. Information intended for manufacturer's use only shall be so designated.

30.1.4 Preferred material reference. Where materials of identical or equal quality can be identified by more than one specification or standard, the drawings shall reference only one such specification or standard. In selecting the specification or standard to be referenced, the following shall be the order of preference:

(a) Federal specification.
(b) Military specification.
(c) Industry and technical society specification or standard.
(d) Manufacturer's specification or standard.

30.1.4.1 Alternate materials drawings. Where materials other than those covered by (a), (b), and (c) of 30.1.4 are referenced, the drawings shall show the complete chemical and physical properties of the proposed material. In addition,

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the drawings shall identify the material in terms of the nearest Federal or Military specification, in order to enable Naval repair facilities to make emergency repairs as necessary from Navy material stocks.

30.1.5 Drawing identification. Contractors who purchase items from subcontractors shall use the subcontractor's drawing number as the single reference identification in all cases where the parts delineated thereon are produced by the subcontractor. The contractor shall not add its drawing number to the drawing, except as an unofficial reference outside the drawing border or margin. Contractors who purchase semi-finished parts from subcontractors for final production phases in their own shops have the option of using, as the single drawing identification, either their own title block and drawing number, or the title block or drawing number of the subcontractor, but not both.

30.1.6 Other components. Where applicable, drawings of drivers and other components shall be in accordance with the specifications covering those equipments.

30.1.7 Tool drawings. Standard complete or limited sets of pullers or parts of puller sets in accordance with GGG-P-781 shall be identified as to type and use in notes on the tool drawings.

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APPENDIX B

MANUAL TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that should be included in technical manuals when required by the contract or order. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

20. APPLICABLE DOCUMENT

20.1 Government document.

20.1.1 Specification. The following specification forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-15071 - Manuals, Technical: Equipments and Systems Content Requirements for.

(Unless otherwise indicated, copies of military specifications are available from the Standardization Documents Order Desk, 700 Robbins Avenue, Bldg. 4D, Philadelphia, PA 19111-5094.)

20.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation.

AMERICAN PETROLEUM INSTITUTE (API)

Manual of Petroleum Standards

(Application for copies should be addressed to the American Petroleum Institute, 1220 L Street, NW, Washington, DC 20005.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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30. MANUAL CONTENTS

30.1 Technical manuals. When required by the contract or order the contractor shall prepare technical manuals in accordance with type I of MIL-M-15071. The purifier manuals shall contain the following features:

(a) A chapter on theory and principles of operation of the purifier.
(b) A description of the theory and use of the different sizes of ring dams, discharge rings, or gravity disk if applicable. This shall include curves showing ring dam or gravity disc selection for the API gravity and specific gravity range differential of the fuel or oil to be purified. The means to adjust the set of gravity discs (see 3.3.10) shall be thoroughly and clearly explained in the technical manual, using a graph to show the adjustment for any specific gravity within the specified limits. The graph or table shall be based on API Manual of Petroleum Standards degrees and specific gravity.
(c) Flow diagrams and recommended installation piping.
(d) A chapter containing a complete description of the purifier equipment. This chapter shall include a discussion of the required purifier routine maintenance procedure, and shall include illustrations of methods to be followed in disassembly, cleaning, and reassembly of the purifier equipment.
(e) A chapter containing a complete description of the purifier control equipment. This chapter shall include a discussion of the required routine maintenance of the controllers, and shall include illustrations of methods to be followed in disassembly, cleaning, and reassembly of the controller equipment.
(f) At least two photographs of the completely assembled purifier unit, taken 180 degrees apart to show as much detail as practicable.
(g) Lubrication requirements.
(h) Utility requirements.
(i) Drawings:

(1) Outline and sectional assembly drawings of motor, controller, and pumps (as applicable).
(2) Purifier sectional drawing with complete parts list.
(3) Purifier outline and installation drawing.
(4) Certification data.

30.1.1 Heater manuals. When oil heaters are required to be furnished with the purifiers, the purifier technical manual shall include all the technical manual information required by the applicable heater specifications. Separate heater manuals will not be required.

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-24710A(SH)	93-06-17

3. DOCUMENT TITLE

PURIFIERS, SOLIDS EJECTING (SELF CLEANING), CENTRIFUGAL, FUEL & LUBE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME TECHNICAL POINT OF CONTACT: TPOC T. MARINO, SEA 05Y23	b. TELEPHONE (Include Area Code) (1) Commercial: (703) 602-7101 (2) AUTOVON: 8-332-7101
c. ADDRESS (Include Zip Code) Commander Naval Sea Systems Command (SEA 05Q42) 2531 Jefferson Davis Hwy Arlington, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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